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Exploring microbial dynamics, metabolic functions and microbes–metabolites correlation in a millennium paddy soil chronosequence using metabolome and microbiome

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Abstract

Background Paddy soil is a typical soil type affected by anthropogenic management and factors related to natural soil formation. The evolution from mudflats to typical paddy soils can significantly affect the soil microecology. Previous studies have reported the evolution of soil physicochemical properties, microbes, and related soil environmental factors in a millennium paddy soil chronosequence. However, the potential biological mechanisms of changes in metabolites and microbes–metabolites interaction are poorly understood. Therefore, a combination of high-throughput sequencing and environmental pseudotargeted metabolomics techniques was adopted to explore the effects of the millennium paddy soil chronosequence on microbial communities, metabolites, and their functions and interactions.

Results The soil ecology changed greatly in the first 60 years of the transition from mudflat to paddy planting. Among the microbial communities, the response of the bacteria to the chronosequence was more sensitive than that of fungi. Among them, the bacterial communities of *Proteobacteria*, *Bacteroidetes*, *Acidobacteria*, and *Nitrospirae* exhibited regular succession over the chronosequence, but the fungal communities did not show regular changes. Bacterial function prediction revealed that the beginning of the critical stage of the evolution from mudflat to paddy soil involved the organic matter cycle and energy flow. In contrast, fungi were characterized mainly by pathogenic and saprophytic functions. The results of the principal component analysis of the metabolites revealed a similar pattern of change as that of the microbes. Seventy-five characteristic metabolites exhibited three trends of change during the development of the paddy soil chronosequence. Twenty-five differentially active metabolic pathways, including glyoxylate and dicarboxylate metabolism, starch and sucrose metabolism, and galactose metabolism, were enriched. In addition, correlation analysis revealed that long-chain fatty acids, short-chain fatty acids, phenolic acids, carbohydrates, and polyalcohols significantly regulate the microbial communities in paddy soil.

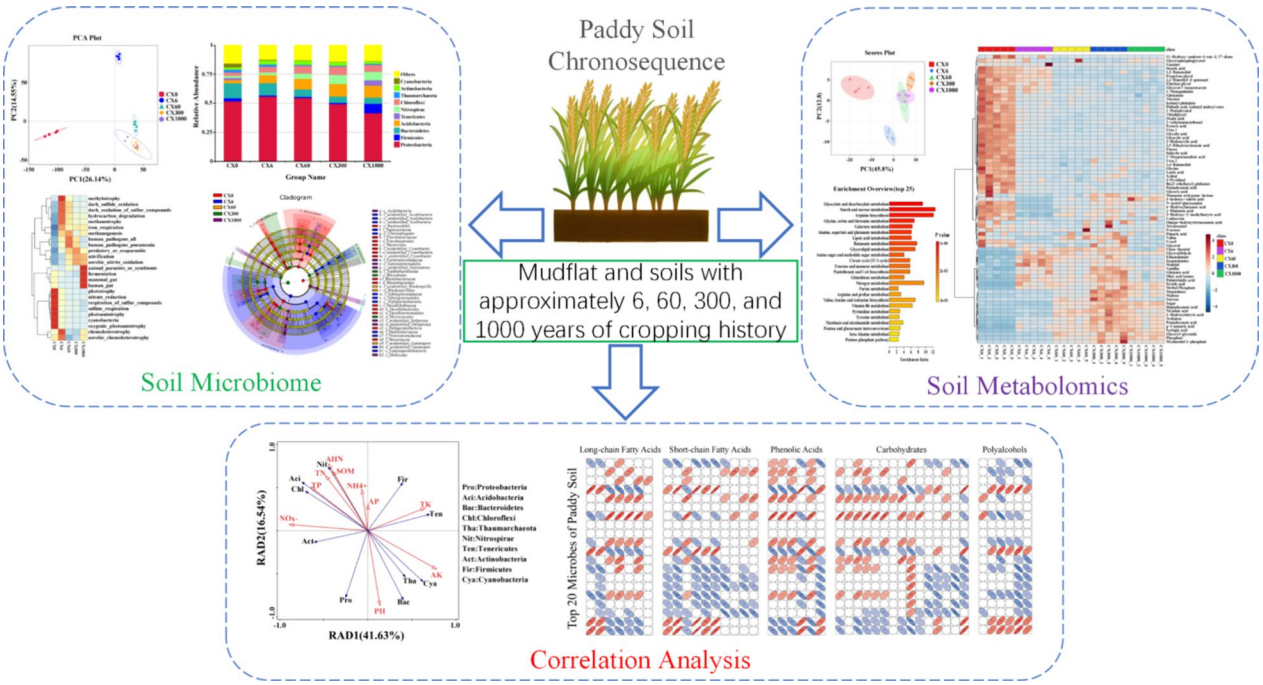
Conclusions Combining metabolome and microbiome has expanded the overall understanding of the development of paddy soil under anthropogenic management. During the development of a paddy soil chronosequence, the synergistic regulation of soil physicochemical properties and metabolites in the microbial community results in increased productivity. This study provides a new perspective on microbes and metabolites interaction.

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Keywords Paddy soil, Microbial community, Metabolic function, Millennium chronosequence, Correlation analysis

Graphical Abstract



Background

As the global population grows, the reclamation of coastal wetlands has become a common and effective practice for alleviating the growing demand for food worldwide [1]. The evolution from mudflats to paddy soils can affect the soil microecology for hundreds or even thousands of years. In the initial stage of development, the soil microbial biomass, nutrient cycling rate, and nutrient availability increase rapidly, after which the gradual phase can last for thousands of years, followed by a possible stage of degradation [2]. Although many potential factors affect soil development and evolution, the influence of the millennium chronosequence far exceeds that of other factors related to soil formation [3]. Paddy soils of different planting years are distributed along the Cixi coastline on the southern bank of Hangzhou Bay, which has a history of reclamation for thousands of years, and approximately 90% of the land in its territory is reclaimed land [4]. Because Cixi paddy soils have very similar parent materials, climates, and biological factors and employ the same paddy planting system, these soils provide valuable materials for studying changes in physicochemical properties, microbes, and metabolites during the development of paddy soil chronosequences [5].

In agroecosystems, soil bacteria and fungi are critical to soil and crop health because of their diverse metabolic functions, and relevant microbes can also promote or inhibit crop and pathogen growth [6–9]. Therefore, soil microbes have always been the core focus, while the role of soil metabolites is often ignored [10]. Metabolites are intermediates or products of enzymatic reactions, including amino acids, sugars, fatty acids, and organic acids, which are closely related to the function, growth, and reproduction of microbes and plants [11]. Moreover, soil metabolites are highly sensitive for discriminating ecosystem productivity [12], and can reflect the occurrence of key biological processes [13]. They can also regulate beneficial microbes in a targeted manner, thereby improving the sustainability of agriculture [14]. For example, *Arabidopsis* produces unique triterpenoid metabolites to assemble and maintain specific microbiota [15]; *N*-acetylglucosamine can mould the community structure and rhizosphere metabolism to promote crop growth [16]; and certain *Pseudomonas* species produce antagonistic compounds that promote induced

resistance to inhibit morbidity in wheat [17]. Therefore, soil metabolites have important implications for predicting different ecological functions and exploring strategies to increase nutrient uptake, soil health, and crop yield through metabolite regulation in agroecosystems [18].

Soil metabolomics has emerged rapidly as a technique for the detection of soil metabolites in recent years. This technology can be used to analyse known and unknown low molecular weight metabolites in soil, which can identify characteristic metabolites and metabolic pathways, and can be combined with microbiome techniques to further identify potential biomarkers and provide functional information on soil microbial responses to abiotic stresses [19]. For example, Li et al. coupled microbial sequencing with soil metabolomics technology and found that biochar and plants jointly regulate soil carbon metabolism and contaminant biodegradation, which is conducive to polycyclic aromatic hydrocarbon degradation by the soil microbiome [20]. Lu et al. demonstrated through the analysis of soil rhizosphere metabolomics and bacterial community structure that long-term cultivation of *Camellia oleifera* can significantly improve soil fertility but may lead to a decline in soil ecological health [21]. Thus, soil metabolomics has been increasingly used to explore different biogeochemical processes in soils and trace metabolic-dependent pathways among soil microbes [22]. Recently, environmental pseudotargeted metabolomics was established for the metabolic profiling of paddy soil chronosequences [12]. However, this study focused mainly on soil metabolomics methods; the interaction between microbes and metabolites needs further study.

Research on the Cixi paddy soil chronosequence has led to major advances in comprehending the patterns and environmental drivers of soil microbial succession under anthropogenic conditions [5, 23, 24]. However, these studies have focussed mainly on soil physicochemical properties and microbial communities without paying attention to metabolites or the relationships between soil metabolites and microbial biomarkers. Therefore, in this study, metabolome and microbiome were integrated to reveal the dynamic changes in microbes and metabolites, metabolic functions, and interactions. The main objectives of this study were to (1) investigate the response of soil properties, microbial communities, and metabolites during the development of a paddy soil chronosequence; (2) explore the interactions between soil physicochemical properties, metabolites, and microbial communities; and (3) analyse the coregulation of the microbial community and metabolites. This study contributes to a comprehensive understanding of long-term succession changes in agricultural soil and promotes the health and sustainable development of paddy agricultural ecosystems.

Materials and methods

Study area

The study area is located in Cixi (30° 02′–30° 24′ N, 121° 02′–121° 42′ E), Zhejiang Province, China, in the eastern Ningshao Plain at the junction of the Shanghai, Hangzhou, and Ningbo triangle regions, with a total area of 1154 km^2. The region has a typical north-subtropical monsoon climate, with a mean annual sunshine duration of 2038 h and a mean annual sunshine percentage of 47%. The mean annual temperature is 16.0 °C, with the highest temperature occurring in July and the lowest occurring in January. Rainfall is abundant, with a mean annual precipitation of 1272.8 mm and a mean annual runoff of 5.122×108 m^3. Cixi has a tradition of building dams to prevent tides, and paddies have long been cultivated. The sampling locations were confirmed on the basis of the historical time of dam construction in Cixi County Annals. Soils chronosequences with 6, 60, 300, and 1000 years of paddy cultivation and mudflat soil were obtained and denoted as CX6, CX60, CX300, CX1000, and CX0 respectively.

Soil sampling

To ensure the comparability of paddy soils of different chronosequences and reduce the influence of fertilization and ploughing, paddy soils were collected before ploughing. Each paddy soil chronosequence had more than 3.3 ha of continuous farmland, and 5 sample points were collected on different farmlands with sampling intervals greater than 100 m. The sample collection process was as follows: a 1 m soil auger was used to sample the topsoil (0–20 cm). To cover the microbes and metabolites in the 0–20 cm soil layer as much as possible, the 0–20 cm soil layer was divided into four parts (0–5 cm, 5–10 cm, 10–15 cm, and 15–20 cm), and equal amounts of samples were taken from each part and mixed evenly.

Five soil samples were collected diagonally for each treatment. The samples were divided into three subsamples for different analyses. For the first subsample, the samples were frozen in liquid nitrogen for 15 min, after which the soil was stored in a dry ice box for microbial sequencing. For the second subsample, samples were collected in a dry ice box, freeze-dried in a vacuum freeze-dryer, and stored at −70 °C for metabolite analysis. In the third subsample, the samples were collected in ice bags and air-dried at room temperature for the determination of soil physicochemical indices.

Soil physicochemical property analysis

A pH meter was used to determine the pH value of the soil solution, and the ratio of soil to aqueous solution (*w/v*) was 1:2.5. The soil total nitrogen (TN) and alkali hydrolysed nitrogen (AHN) contents were determined

via a Kjeldahl nitrogen analyser and a continuous flow analyser using the potassium dichromate-sulfuric acid digestion and wet alkali digestion methods, respectively. The total phosphorus (TP) and total potassium (TK) contents in the soil were determined via a spectrophotometer and atomic absorption spectroscopy using the sodium hydroxide melting method and the molybdenum blue method. Soil-available phosphorus (AP) and potassium (AK) were extracted with 0.5 M $NaHCO_3$ (pH 8.5) at a ratio of 1:20 (*w/v*) and 1 M CH_3COONH_4 (pH 7.0) at a ratio of 1:10 (*w/v*), respectively, and then determined via a continuous flow analyser and atomic absorption spectrometry. The soil organic matter (SOM) was extracted with 0.4 M potassium dichromate and a sulfuric acid solution at a ratio of 1:100 (*w/v*) and determined via the titration method. Ammonium nitrogen (NH_4^+) and nitrate nitrogen (NO_x^-) [i.e., nitrite (NO_2^-) and nitrate (NO_3^-)] were extracted with 2 M KCl at a ratio of 1:5 (*w/v*) and determined via a continuous flow analyser at wavelengths of 660 nm and 550 nm. All physicochemical properties were determined according to Chinese agricultural standard methods.

Soil DNA extraction, library construction and high-throughput sequencing

The genomic DNA of the samples was extracted via CTAB, and then the DNA purity and concentration were determined via agarose gel electrophoresis to evaluate the purity and integrity of the soil DNA. An appropriate amount of sample DNA was diluted to 1 ng/μl in a centrifuge tube with sterile water. Using the diluted genomic DNA as a template, PCR was performed via specific primers with Barcode and Phusion® High-Fidelity PCR Master Mix with GC Buffer (New England Biolabs, China) to ensure amplification efficiency and accuracy. The primers 515F (5′-GTGCCAGCMGCC GCGGTAA-3′) and 806R (5′-GGACTACH VGGGTWT CTAAT-3′) were used to amplify the V3 − V4 region of the bacterial 16S rRNA gene to characterize bacterial diversity. The primers ITS5-1737F (5′-GGAAGTAAA AGTCGTAACAAGG-3′) and ITS2-2043R (5′-GCTGCG TTCTTC ATCGATGC-3′) were used to amplify the internal transcribed spacer region 1 (ITS1) of the fungi to characterize fungal diversity.

The PCR products were visualized via 2% agarose gel electrophoresis. The qualified PCR products were subsequently purified with magnetic beads and quantified via enzyme labelling. Equal amounts of samples were mixed according to the concentration of the PCR products. After full mixing, the mixed PCR products were subjected to electrophoresis on a 2% agarose gel for quality assessment and purified with the GeneJETTM Gel Extraction Kit (Thermo Scientific, China). The library was constructed by using the Ion Plus Fragment Library Kit (Thermo Fisher, China). After the constructed library was qualified by Qubit quantification and library detection, on-machine sequencing was performed via an Ion S5™ XL instrument (Thermo Fisher, China). The raw data were deposited in the NCBI Sequence Read Archive (SRA) database (accession numbers: PRJNA1087732 and PRJNA1088256).

Soil environmental pseudotargeted metabolomics analysis

Environmental pseudotargeted metabolomics was applied for metabolite analysis. Briefly, soil metabolites were sequentially extracted with 1.5 mL of methanol:water (*v:v*, 3:2) twice and 0.8 mL of water. The extract solution was freeze-dried and then subjected to microwave-assisted derivatization. Seventeen microlitres of methoxyamine hydrochloride (25 mg mL^{-1}) was used for oximation, and then, 103 μL of BSTFA (1% TMCS) was added for silylation. The oximated sample (3 + 2 min) and the silylated sample (3 + 3 + 3 min) were reacted with 800 W of microwave energy. The supernatant was analysed via GC–MS with a 1-μL injection. The identification and quantification parameters are described in the literature [12].

Data analysis

The observed OTU, Chao1, Shannon, and Simpson indices were calculated via QIIME software (version 1.9.1), and graphs related to the microbes were drawn via R software (version 2.15.3) and Origin 2018 software. Principal component analysis (PCA) of the metabolites was performed via MetaboAnalyst 5.0. The metabolites of 0- and 6-year, 6- and 60-year, and 0- and 60-year paddy soils were analysed via orthogonal partial least squares discriminant analysis (OPLS-DA). Metabolites with a *t* test p value < 0.05 and a variable importance in projection (VIP) value > 1.2 were defined as characteristic metabolites. The characteristic metabolites were collected to construct a clustering heatmap and analyse the biological pathways. Redundancy analysis (RDA) of the soil physicochemical properties and microbes was performed via CANOCO5. The correlation heatmaps of characteristic metabolites and microbes were generated via Origin 2018.

Results

Soil physicochemical properties

In the millennium paddy soil chronosequence, the measured values of the 10 soil physicochemical indices showed regular and gradual changes (Table 1). With increasing planting time, the soil pH decreased significantly from 8.04 under CX0 to 6.84 under CX1000. In contrast, AHN increased with increasing paddy soil chronosequence. In short-term (0–60 years) reclamation, AP, TP, SOM,

Table 1 Changes of physical and chemical properties of millennium paddy soil chronosequence

Paddy soil	pH (KCl)	AHN (mg kg^{-1})	AP (mg kg^{-1})	AK (mg kg^{-1})	NH_4^+ (mg kg^{-1})	NO_x^- (mg kg^{-1})	TN (%)	TP (%)	TK (%)	OM (g kg^{-1})
CX0	8.04 ± 0.02a	38.13 ± 0.96d	17.93 ± 0.60bc	467.49 ± 8.23a	5.72 ± 0.87b	3.22 ± 0.24c	0.0544 ± 0.0014c	0.0442 ± 0.0197d	1.61 ± 0.04a	10.73 ± 0.47b
CX6	7.88 ± 0.12b	58.51 ± 3.50c	13.24 ± 1.44c	72.87 ± 13.19d	3.71 ± 0.20c	44.37 ± 4.77a	0.0384 ± 0.0020d	0.0647 ± 0.0004c	1.37 ± 0.02c	8.39 ± 1.50b
CX60	7.84 ± 0.05b	96.89 ± 15.69b	30.29 ± 6.73a	147.64 ± 22.13b	7.26 ± 1.52a	42.83 ± 10.86a	0.1749 ± 0.0163a	0.1058 ± 0.0057a	1.51 ± 0.02b	33.16 ± 5.66a
CX300	7.32 ± 0.13c	133.08 ± 12.57a	14.08 ± 2.44c	94.02 ± 2.95c	6.73 ± 0.54ab	42.49 ± 2.58a	0.1773 ± 0.0113a	0.0964 ± 0.0167a	1.51 ± 0.04b	29.11 ± 2.94a
CX1000	6.84 ± 0.05d	139.56 ± 12.24a	20.63 ± 2.74b	79.32 ± 5.99 cd	5.88 ± 0.19b	24.65 ± 1.74b	0.1514 ± 0.0125b	0.0804 ± 0.0030b	1.54 ± 0.02b	30.71 ± 0.92a

ANH alkali hydrolyzed nitrogen, *AP* available phosphorus, *AK* available potassium, *TN* total nitrogen, *TP* total phosphorus, *TK* total potassium, *OM* organic matter

and NH_4^+ decreased but subsequently increased and reached their maximum values in CX60. Then, these indices changed slowly from CX60–CX300–CX1000. The changes in TN were similar, but TN reached its maximum in CX300. Overall, 60 years of reclamation may be an important turning point in terms of physicochemical properties in paddy soil ecosystems.

Soil microbial community

In the first 60 years, there were significant changes in bacterial diversity (Table 2). This is clearly shown by the Shannon index, which peaked at CX60. Consistent with these findings, the number of bacterial species and bacterial richness also increased to their maximum values during the same period. These indices subsequently decreased in CX60–CX300–CX1000. In contrast, bacterial evenness did not show a similar pattern. The change in the fungal diversity index over the millennium paddy soil chronosequence was slightly different from that of bacteria. The fungal Shannon index and evenness increased after a significant decrease from CX0 to CX6 and reached a maximum value in CX300.

PCA was further used to evaluate the changes in the microbial community during the development of the paddy soil chronosequence (Fig. 1A, C). Compared with that of the mudflat (CX0), the aggregation position of the paddy soil changed significantly between CX6 and CX60–CX300–CX1000. The influence of the chronosequence on the bacterial community composition was significantly greater than that on the fungal community composition. At the phylum level, the response of the top 10 bacterial communities to the chronosequence was very obvious (Fig. 1B, D). *Proteobacteria* was the most abundant bacterial phylum in the paddy soil chronosequence; its abundance first increased in CX0–CX6 but then decreased. The abundances of *Bacteroidetes* and *Cyanobacteria* decreased continuously with increasing planting time, from 12.50 to 5.30% and from 3.22 to 0.24%, respectively. The abundances of *Acidobacteria*, *Nitrospirae*, and *Chloroflexi* increased significantly with the chronosequence and reached their maximum values under CX300, from 2.78 to 10.78%, from 1.88 to 7.73% and from 2.91 to 7.23%, respectively. Compared with that of the bacterial community, succession pattern of the fungal community was not obvious.

A logarithmic linear discriminant analysis (LDA) score threshold of 4.0 was used as a discriminant feature for LEfSe analysis to identify specific bacterial and fungal taxa (Figs. S1 and S2). Fifty-three rich bacterial branches (19 in CX0, 13 in CX6, 4 in CX60, 7 in CX300, and 10 in CX1000) and 35 rich fungal branches (10 in CX0, 5 in CX6, 4 in CX60, 10 in CX300, and 6 in CX1000) were identified. Figure 2 shows the taxonomic information of key bacterial and fungal types in the paddy soil chronosequence, and dramatic differences in the species were annotated and analysed. *Cyanobacteria*, *Bacteroidetes*, *Chloroflexi*, *Nitrospirae*, *Acidobacteria*, and *Proteobacteria* in the bacterial communities and *Mortierellomycota* in the fungal communities were the main microbial biomarkers at the phylum level.

FAPROTAX and FUNGUILD function predictions were used to establish clustered heatmaps on the basis of the functional abundances of bacteria and fungi, respectively (Fig. 3A, B). The bacterial function prediction revealed that the mudflat is enriched in terms of sulfur compound respiration, sulfate respiration, photoautotrophy, oxygenic photoautotrophy, and nitrate reduction. CX6 is enriched in methylotrophy, methanotrophy, dark sulfide oxidation, chemoheterotrophy, aerobic chemoheterotrophy, and hydrocarbon degradation. CX60–CX300–CX1000 are enriched in aerobic nitrite oxidation, nitrification, and predatory or exoparasitic processes. The fungal function prediction revealed that the mudflat is enriched in ectomycorrhizal, animal pathogen, lichenized, and fungal parasites. CX6 is enriched in endophyte–plant pathogens and plant pathogen–undefined saprotrophs. CX60–CX300–CX1000 are enriched mainly in dung saprotrophs, leaf saprotrophs, animal pathogens, soil saprotrophs, and arbuscular mycorrhizae.

Table 2 Microbial community alpha diversity index in millennium paddy soil chronosequence

Paddy soil	Observed species		Simpson index		Shannon index		Chao1 estimator	
	B	F	B	F	B	F	B	F
CX0	3207 ± 96b	611 ± 64b	0.9956 ± 0.0005ab	0.9648 ± 0.0097a	9.58 ± 0.09ab	6.30 ± 0.34ab	3552 ± 58b	810 ± 72b
CX6	3384 ± 149ab	735 ± 35ab	0.9972 ± 0.0004a	0.8658 ± 0.0376b	10.05 ± 0.07a	5.23 ± 0.42b	3808 ± 311ab	962 ± 67ab
CX60	3638 ± 127a	888 ± 239a	0.9968 ± 0.0007a	0.8942 ± 0.1315ab	10.14 ± 0.08a	6.29 ± 1.47ab	4134 ± 269a	1135 ± 300a
CX300	3424 ± 195ab	885 ± 173a	0.9974 ± 0.0005a	0.9822 ± 0.0052a	10.09 ± 0.08a	7.46 ± 0.35a	3864 ± 426ab	1044 ± 247ab
CX1000	3279 ± 360b	875 ± 127a	0.9896 ± 0.0095b	0.971 ± 0.0180a	9.46 ± 0.78b	7.13 ± 0.68a	3758 ± 401ab	1076 ± 127ab

B bacteria, *F* fungi

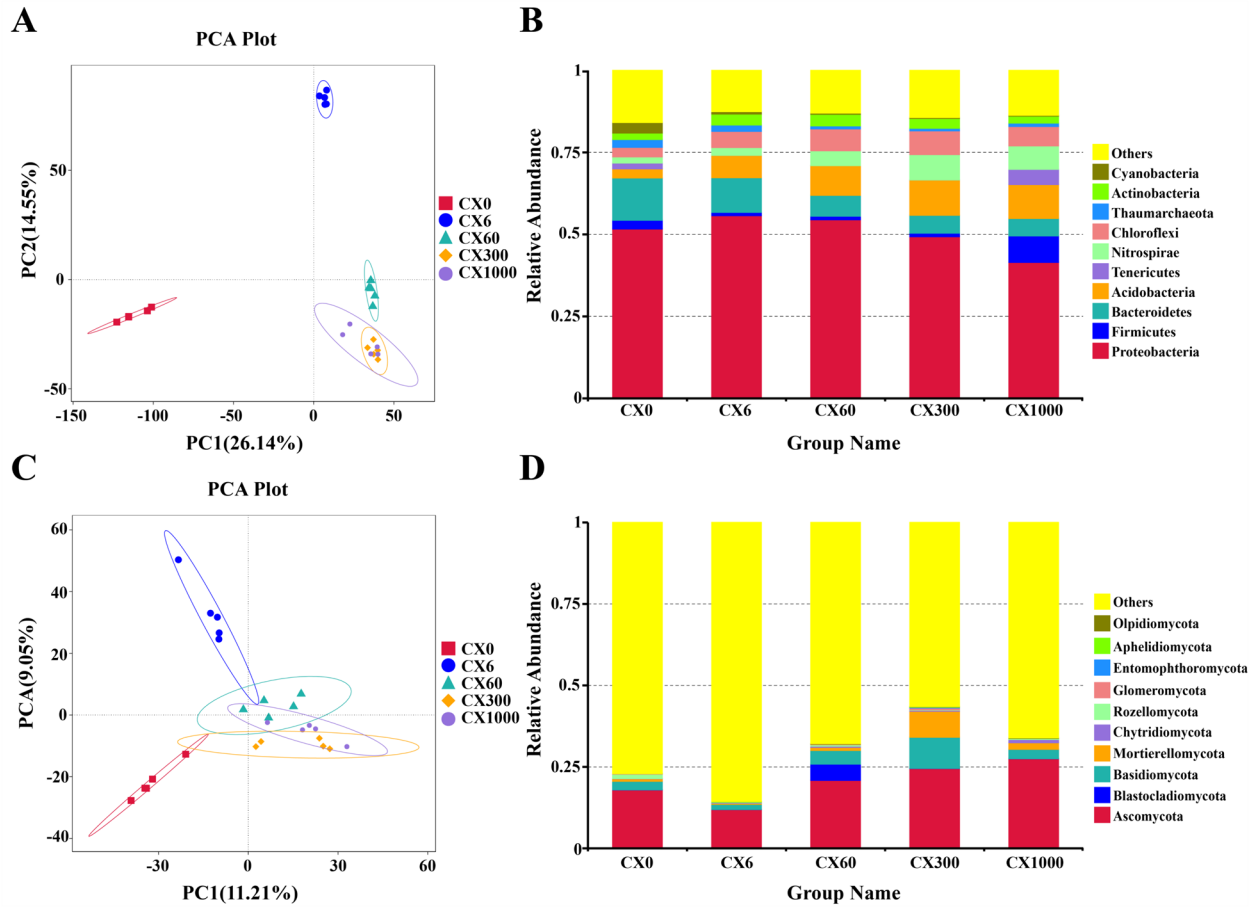


Fig. 1 Principal component analysis (PCA) of soil bacterial (**A**) and fungal (**C**) communities in paddy soil chronosequences on the basis of Euclidean distances. Relative abundance histograms of the bacterial (**B**) and fungal (**D**) community structures in the paddy soil chronosequence at the phylum level

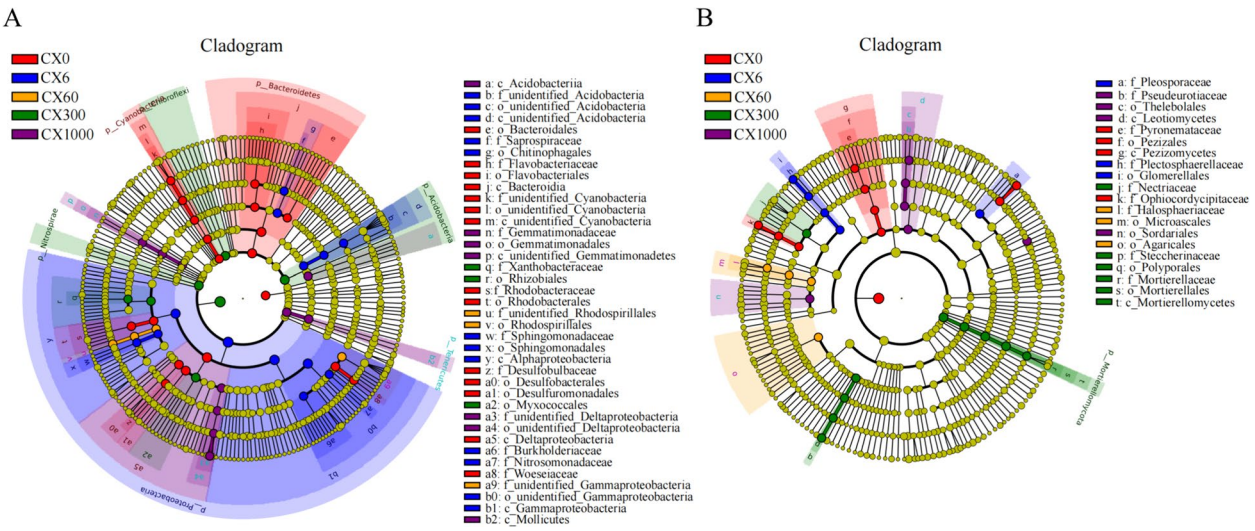


Fig. 2 Key bacterial (**A**) and fungal (**B**) communities in response to the paddy soil chronosequence obtained via LEfSe analysis; an LDA significance threshold of > 4.0 is shown

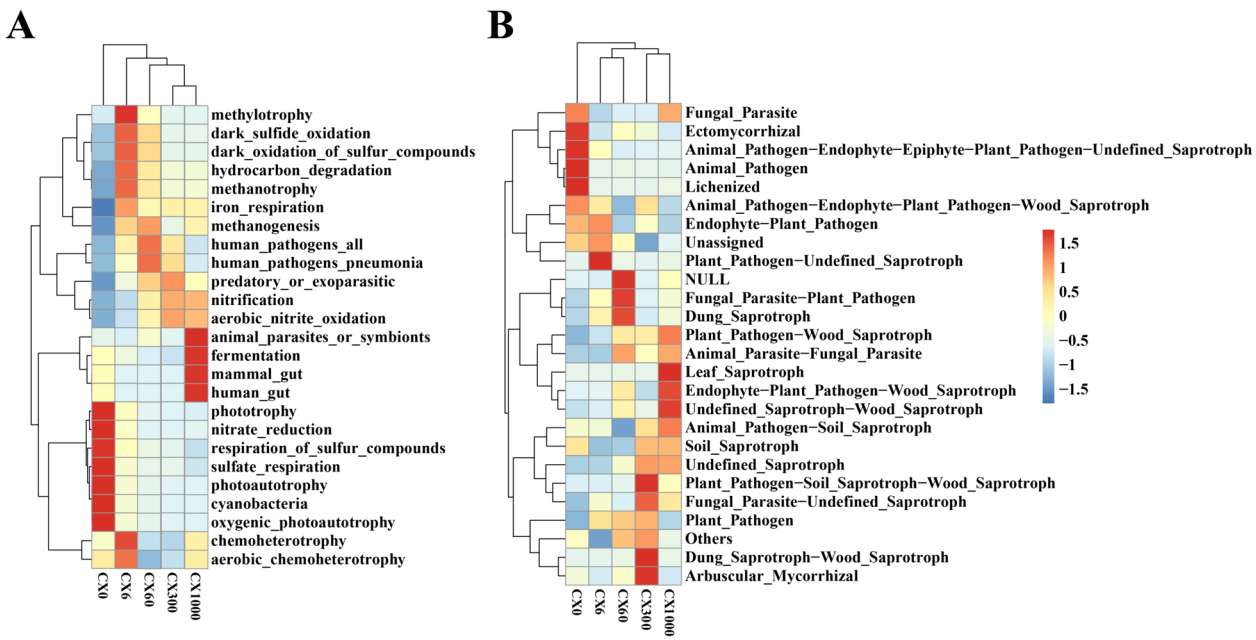


Fig. 3 Distribution of the functional prediction of soil bacteria (**A**) and fungi (**B**) in the paddy soil chronosequence with the FAPROTAX and FUNGUILD databases (Different colours indicate the relative abundance of a single sample, with red representing functions with higher abundance and blue representing functions with lower abundance)

Soil metabolites

PCA of the metabolites revealed that the metabolite compositions of CX0 and CX6 are significantly different from those of CX60–CX300–CX1000. The dynamic change in metabolites in the late stage of evolution (CX60–CX300–CX1000) is small, so three soil chronosequences, CX0, CX6, and CX60, were selected to screen characteristic metabolites, and a total of 75 characteristic metabolites were screened (Fig. 4A).

Heatmap analysis revealed the change trends of 75 characteristic metabolites in the millennium paddy soil chronosequence, and the metabolites could be classified into three groups (Fig. 4B). The first group decreased in abundance significantly in CX0–CX6, then fluctuated with the chronosequence, and consisted mainly of the short-chain fatty acids pyruvic acid, 3-hydracrylic acid, lactic acid, glyoxylic acid, glycolic acid, and polyalcohols 1,4-butanediol, propylene glycol, ethylene glycol, thiodiglycol, and 1,3-butanediol. The second group decreased in abundance significantly in CX0–CX6, followed by an increase in CX6–CX300 and a subsequent decrease with the chronosequence, and consisted mainly of the short-chain fatty acids 3-hydroxy-valeric acid, 4-hydroxybutanoic acid, 2-pentenoic acid, and 3-hydroxy-2-methylbutyric acid. The third group increased in abundance from CX0–CX6–CX60–CX300 and decreased in abundance from CX300–CX1000 and mainly consisted of the long-chain fatty acids oleic acid isomer, palmitelaidic acid, heptadecanoic acid isomer, and pentadecanoic acid branched-chain and the phenolic acids vanillin, *p*-coumaric acid, ferulic acid, and syringic acid.

The top 25 differential metabolic pathways were identified via enrichment analysis of 75 characteristic metabolites, as shown in Fig. 4C. The first three metabolic pathways are glyoxylate and dicarboxylate metabolism, arginine biosynthesis, and starch and sucrose metabolism. Moreover, other metabolic pathways, including alanine, aspartate and glutamate metabolism; glycine, serine and threonine metabolism; galactose metabolism; lipoic acid metabolism; and butanoate metabolism, were also significantly enriched. These metabolic pathways are key pathways for carbon and nitrogen in soil microbes.

Correlations between microbes and environmental factors or metabolites

RDA revealed that the soil physicochemical properties significantly affected the microbial community structure (Fig. 5). In the bacterial communities, *Proteobacteria* was most significantly affected by pH compared with the other soil physicochemical properties. TK and AK were the main physicochemical properties affecting *Tenericutes* and were positively correlated with *Tenericutes*. The abundances of *Bacteroidetes*, *Thaumarchaeota*, and *Cyanobacteria* were positively correlated with pH, TK, and AK but negatively correlated with other

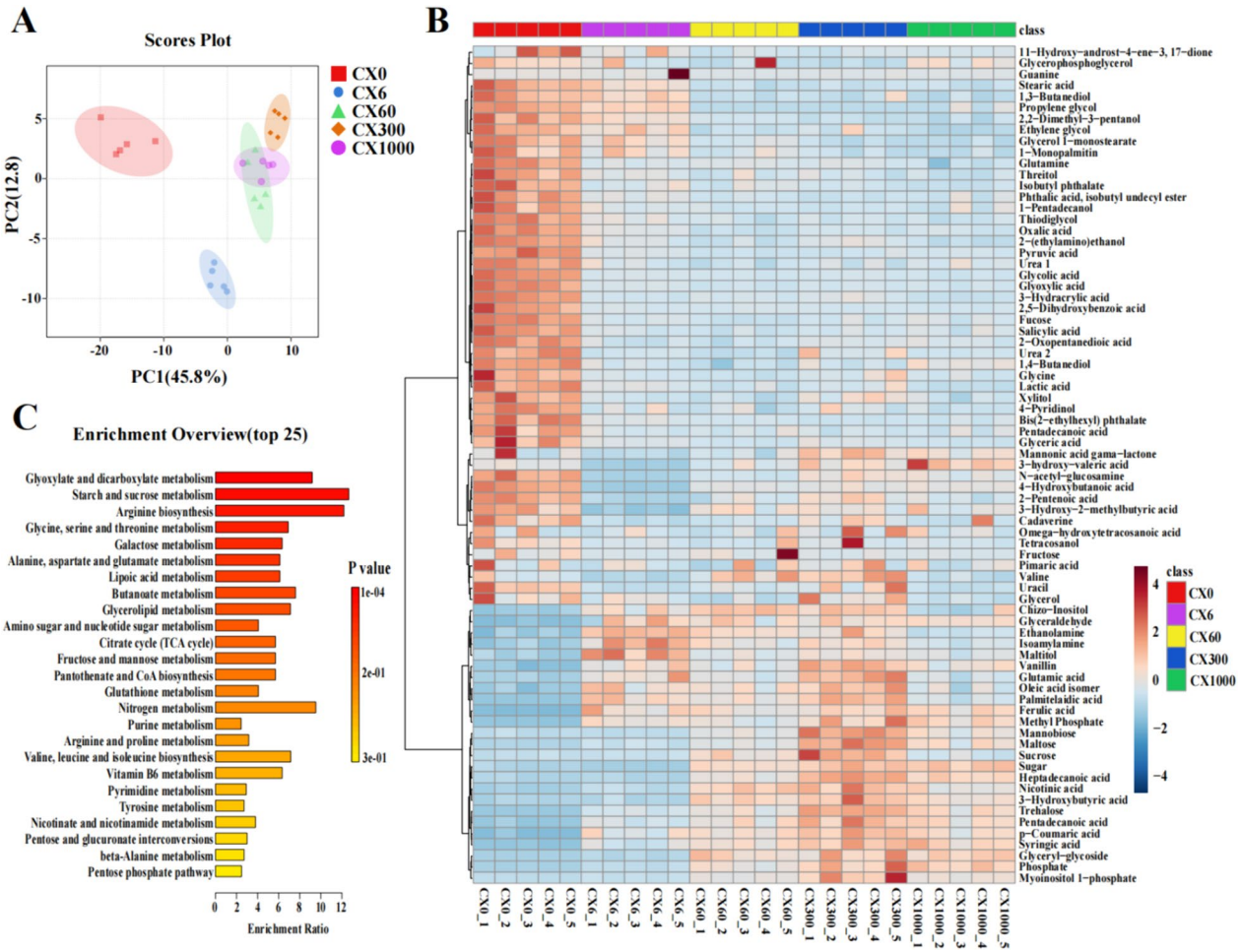


Fig. 4 **A** Metabolic difference analysis of the paddy soil chronosequence via principal component analysis (PCA). **B** Heatmap clustering of 75 characteristic metabolites (VIP > 1.2 and $p < 0.05$). **C** Enrichment histogram of metabolic pathways with characteristic metabolites

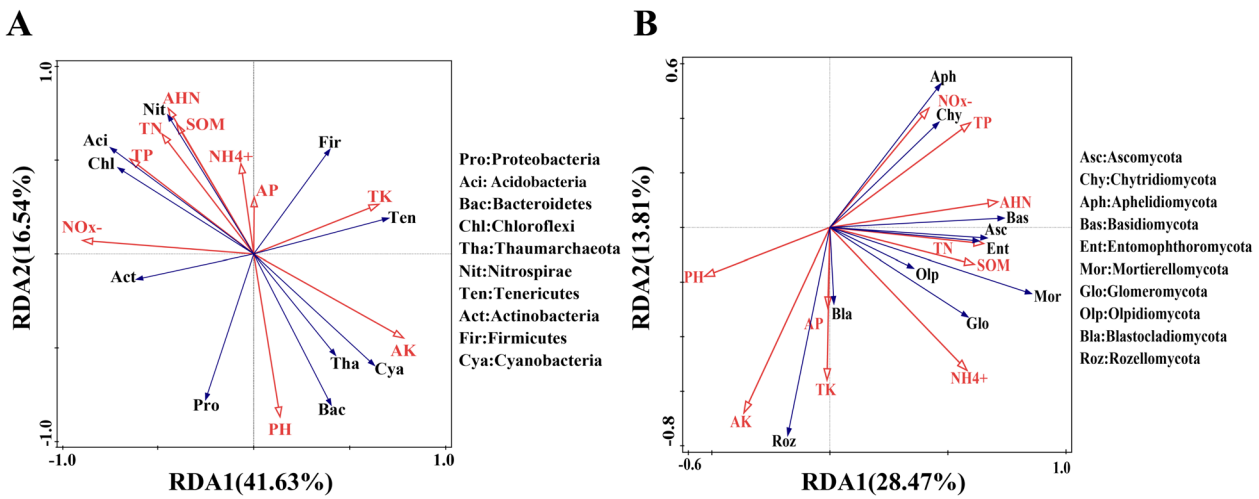


Fig. 5 Redundancy analysis (RDA) to identify the relationships between the abundances of bacterial (**A**) fungal (**B**) phylum-level taxa (blue arrows) and soil physicochemical properties (red arrows)

physicochemical properties. In contrast, *Nitrospirae*, *Acidobacteria*, and *Chloroflexi* were negatively correlated with pH, TK, and AK but positively correlated with other physicochemical properties. The fungal communities had a lower correlation with soil physicochemical properties than the bacterial communities did, and the correlations were mainly positive.

An analysis of the interactions between metabolites and microbes revealed that the correlations between metabolites and bacterial genera were significantly greater than those between metabolites and fungal genera (Figs. 6 and 7). The characteristic metabolites associated with microbes are mainly long-chain fatty acids, short-chain fatty acids, phenolic acids, polyalcohols, and carbohydrates. In the bacterial communities, the long-chain fatty acids pentadecanoic acid and stearic acid were negatively correlated with most bacterial genera, whereas pentadecanoic acid branched-chain, palmitelaidic acid, and oleic acid isomer were positively correlated with most bacterial genera. With the exception of 3-hydroxybutyric acid, there were mainly negative correlations between short-chain fatty acids and bacterial genera. Among the phenolic acids, 2,5-dihydroxybenzoic acid and salicylic acid were mainly negatively correlated with the bacterial genera, whereas syringic acid, ferulic acid, vanillin, and *p*-coumaric acid were mainly positively correlated with the bacterial genera. In addition, carbohydrates and polyalcohols were negatively correlated with bacterial genera. In the fungal communities, certain long-chain fatty acids, short-chain fatty acids, phenolic acids, carbohydrates, and polyalcohols, such as oleic acid isomer, syringic acid, ferulic acid, *p*-coumaric acid, and vanillin, were positively correlated with fungal genera.

Discussion

Response of soil physicochemical properties to the chronosequence

The soil physicochemical properties were closely related to the chronosequence of paddy soil reclamation. Continuous irrigation and artificial cultivation and fertilization led to the desalination and maturation of the mudflat. In particular, in the first 60 years, the accumulation of coastal sediments [23], the low decomposition rate of SOM under flooding conditions, and the input of paddy straw and microbes promoted the accumulation of TN and SOM [25]. Compared with that in the mudflat, the TP content significantly increased in the first 60 years, which was related to the long-term application of phosphate fertilizer, whereas the reduction in the TP content in the later period may be related to the loss of phosphorus adsorbents [26]. The soil pH continued to decline during the development of the paddy soil chronosequence, which was partly due to soil acidification caused by nitrogen fertilizer application [27]. Moreover, the decomposition of SOM under flooding management and the leaching effect of irrigation may also explain the decrease in paddy soil pH [28], both of which increase the proton concentration. All these soil physicochemical properties are important characteristics during the development of a paddy soil chronosequence, confirming that the soil has developed in chronological order.

Response of soil microbes to the chronosequence

Microbes constitute the core of the ecological functions of the soil and are considered early and sensitive indicators of changes in soil quality [6]. Paddy topsoil is often in a reduced state due to water saturation. In the process of hydroponic ripening under hypoxic conditions, the accumulation of SOM and potential nutrients increases. Hence, the diversity index of the soil bacterial community rapidly increased in the first 60 years [29]. Under the long-term influence of such hydrothermal conditions and substrates, specific microbial community structures and functional activities are established. As a result of niche competition [30] and intensive agricultural management [31], these indicators fluctuated and declined in the following centuries after 60 years, and the overall changes were smaller than those in the first 60 years. The number and richness of the soil fungal communities were consistent with the changes in the bacterial communities, but the overall diversity and evenness significantly decreased in the first 6 years, probably due to artificial submergence [32].

A comparison of the PCA results for bacteria and fungi (Fig. 1A, C) revealed that bacteria were more sensitive to the chronosequence than were fungi. Generally, fungi are more sensitive to depth, possibly because depth affects oxidation–reduction conditions [32]. Growth conditions advantageous to fungi are present in the more aerated surface of the paddy soil, but the chronosequence represents the comprehensive changes in various physicochemical properties. The abundance of *Proteobacteria* decreased from CX6–CX1000, possibly because the genetically inherited *Deltaproteobacteria*, *Gammaproteobacteria*, and *Alphaproteobacteria* in seawater are less competitive than other bacteria in agricultural environments [33–35]. The abundance of *Bacteroidetes* continued to decline over the chronosequence, mainly because some species such as *Bacteroidia* and *Flavobacteriales* in *Bacteroidetes,* are dominant under high salt stress [36]. Long-term fertilization caused a decrease in pH, which led to an increase in *Acidobacteria,* which have a preference for acidity [37]. *Chloroflexi* are related to the soil water content and usually grow under strict anaerobic conditions; consequently, the abundance of *Chloroflexi* increased over the paddy soil chronosequence [35].

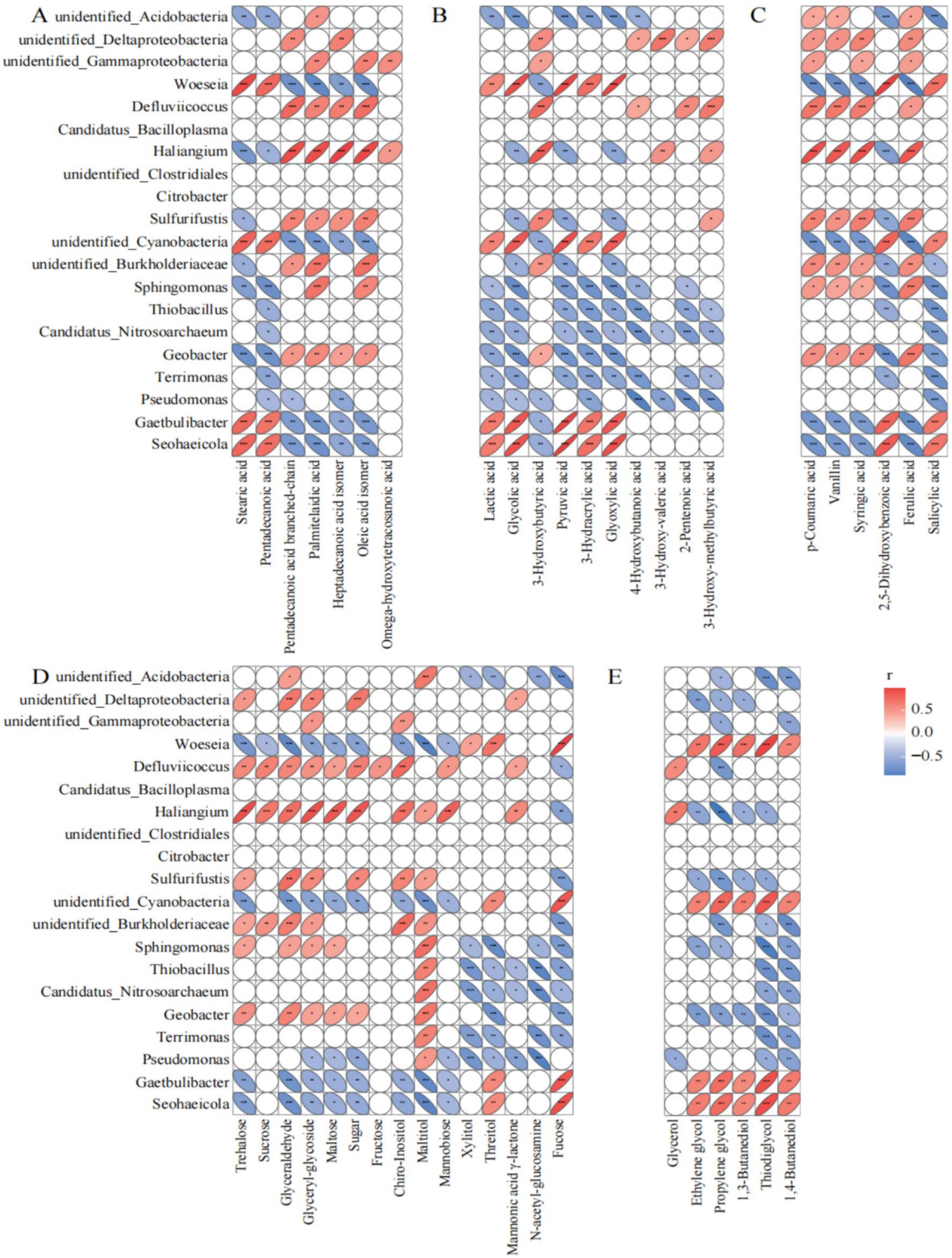


Fig. 6 Correlation heatmaps between characteristic metabolites and the top 20 bacterial groups: **A** long-chain fatty acids and bacterial genera; **B** short-chain fatty acids and bacterial genera; **C** phenolic acids and bacterial genera; **D** carbohydrates and bacterial genera; **E** polyalcohols and bacterial genera. The size of the ellipse area represents the size of the correlation, and the different directions of the ellipse represent positive or negative correlations ($p \le 0.001$, ***; $0.001 < p \le 0.01$, **; $0.01 < p < 0.05$, *; $p \ge 0.05$, no asterisk)

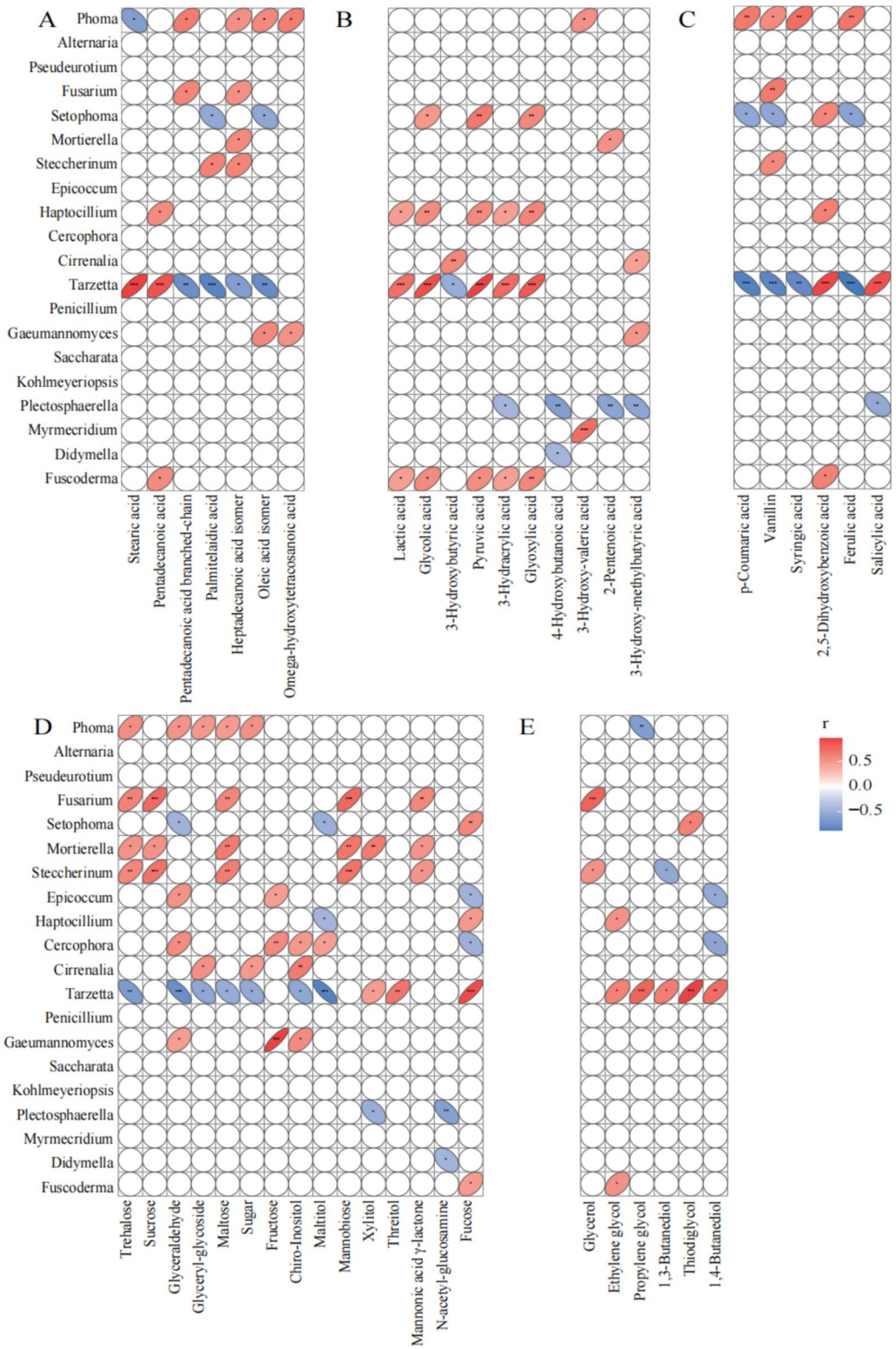


Fig. 7 Correlation heatmaps between characteristic metabolites and the top 20 fungal groups: **A** long-chain fatty acids and fungal genera; **B** short-chain fatty acids and fungal genera; **C** phenolic acids and fungal genera; **D** carbohydrates and fungal genera; **E** polyalcohols and fungal genera. The size of the ellipse area represents the size of the correlation, and the different directions of the ellipse represent positive or negative correlations ($p \le 0.001$, ***; $0.001 < p \le 0.01$, **; $0.01 < p < 0.05$, *; $p \ge 0.05$, no asterisk)

Among the fungal communities, *Ascomycota* was the key microbial driving force and overwhelmingly dominated the paddy soil chronosequence; these fungi can rapidly decompose and utilize complex organic compounds and promote the conversion of mudflat soil to paddy soil [32]. Furthermore, *Mortierellomycota* can decompose lignin and cellulose, and its mycelia can change soil microhabitats by affecting soil aggregates [38]. The succession pattern of the fungal community structure was not as obvious as that of the bacterial community structure, but the fungal community structure played an important role in paddy soil. Fungi can decompose SOM that is difficult to degrade, and the products can then be utilized by bacteria [39].

In bacterial function prediction, chemoheterotrophy and aerobic chemoheterotrophy are closely related to the circulation of organic matter and the flow of energy in the soil system [40], which was significantly enriched in CX6. These results indicated that the initial stage of evolution was the main stage of the organic matter cycle and energy flow. Nitrification and aerobic nitrite oxidation were enriched in CX60–CX300–CX1000, indicating that long-term inorganic nitrogen application promoted nitrogen oxidation and conversion to nitrate [41]. With respect to fungal function prediction, owing to the long-term anaerobic state of paddy soil, increased soil stability and ecosystem self-purification capacity [42], the abundance of pathogens gradually decreased, especially after 60 years of paddy cultivation. However, the abundance of saprophytic bacteria increased significantly, which was mainly related to the increase in the relative abundance of *Ascomycetes* [43]. The changes in the microbial diversity index, community structure, and functional prediction with respect to the paddy soil chronosequence indicated that microbial changes under long-term agricultural activities were related to soil evolution, which was ultimately driven by soil physicochemical properties.

Response of soil metabolites to the chronosequence

On the basis of GC–MS environmental pseudotargeted metabolomics analysis, the content and composition of small-molecule metabolites significantly changed over the paddy soil chronosequence. The metabolite compositions of CX0 and CX6 significantly differed from that of CX60–CX300–CX1000. The changes in soil metabolites and microbes were closely related. Soil microbes mainly obtain metabolites from soil organic matter and its biomass turnover, and the metabolic activities of microbes feed back into the metabolite bank [44]. Therefore, the microbes and metabolites presented similar patterns across the paddy soil chronosequence.

The characteristic metabolites of the paddy soil chronosequence showed three different trends of change, and the main metabolites were long-chain fatty acids, short-chain fatty acids, phenolic acids, and carbohydrates. The main components of phospholipids in cell membranes are long-chain fatty acids, which are closely related to the growth and reproduction of microbes [45]. The long-chain fatty acids oleic acid isomer and palmitic acid promote the transformation of soil organic nutrients and stimulate improvements in soil enzyme activity [46]. Compared with that in CX6, the enrichment of short-chain fatty acids such as 3-hydroxyvaleric acid, 2-pentenoic acid, and 4-hydroxybutanoic acid in CX60–CX300–CX1000 inhibited the growth and spread of certain soil pathogens and parasites [47]. The contents of phenolic compounds such as ferulic acid, syringic acid, and *p*-coumaric acid in CX60–CX300–CX1000 were significantly greater than those in CX0–CX6, which was caused by the long-term low-oxygen state in the paddy soil. Low oxygen inhibits the activity of phenoloxidase and promotes the accumulation of phenolic substances [48]. The contents of carbohydrates such as trehalose, sugar, sucrose, and maltose significantly increased in CX60–CX300–CX1000. As sensitive indices for measuring cold resistance, they can protect plant biofilms and maintain osmotic pressure, thus increasing plant tolerance to environmental stress, which is highly beneficial for paddy planting [12].

The enrichment analysis revealed that the characteristic metabolites were enriched mainly in amino acid metabolism, energy metabolism, and carbohydrate metabolism. Starch and sucrose metabolism are the most significantly varied metabolic pathways in the paddy soil chronosequence, which is attributed to sucrose being an important bacterial carbon source for energy storage in SOM-rich environments [12]. Galactose metabolism and lipoic acid metabolism also changed markedly. These carbohydrate metabolic pathways are closely related to nitrogen fixation and phosphorus dissolution. These effects are conducive to promoting nitrogen and phosphorus cycling in soil and preventing nitrogen and phosphorus loss from fertilizer [21]. The intensification of glyoxylic acid and dicarboxylic acid metabolism can reduce the production of NADH and then lead to a decrease in hydroxyl radical levels. However, the glycolic acid, oxalic acid, and glycine contents significantly decreased after paddy planting, which may have resulted in an increase in hydroxyl radical levels [49]. The long-term application of nitrogen fertilizer and organic fertilizer in paddy soil affects carbon and nitrogen metabolism-related pathways, such as arginine biosynthesis; glycine, serine and threonine metabolism; and alanine, aspartate and glutamate metabolism [2]. These changes in amino acid metabolism are caused by the survival of microbes in response to the development of the paddy soil chronosequence and can help

microbes absorb amino acids, accelerate soil organic mineralization, and promote the utilization of nitrogen by plants [50]. These research results showed that the main role of metabolic functional genes is to ensure the growth of microbes through the intake of amino acids, carbohydrates, vitamins, and other nutrients.

Relative stability of the microbial community structure and metabolite composition from the mudflat to the paddy soil took approximately 60 years to develop in this study. Cui et al. [24] reported that these evolutionary processes took only approximately 40 years in the paddy soil chronosequence on Chongming Island. Paddy soil evolution is affected mainly by the duration of reclamation, followed by agricultural management, such as fertilization, tillage measures, and tillage systems [29]. Modern intensive agricultural management can quickly drive and regulate the nutrient, water, and energy cycles in the soil, accelerate the conversion of coastal wetlands to farmland, and thus improve ecosystem productivity [51]. This rapid evolution has also resulted in adverse effects, such as accelerated soil acidification, increased water and soil pollution, eutrophication of coastal ecosystems, reduced biodiversity, and reduced soil sustainability. To reduce the risks associated with reclamation, ecological improvement measures can be implemented, such as applying more organic fertilizer and returning straw to the field to improve soil fertility [52]. Buffer zones of grass and forest can also be established in reclamation areas and along land borders to reduce the risk of nitrogen and phosphorus erosion [53].

Physicochemical properties and metabolites codetermine the microbial community in a paddy soil chronosequence

In the bacterial communities, *Proteobacteria* was positively correlated with pH, and the relative abundance of *Proteobacteria* decreased with decreasing pH in CX6–CX60–CX300–CX1000, consistent with the results of Zhalnina et al. [37]. Li et al. [54] explained the positive correlation between *Nitrospirae* and nitrogen by reviewing the progress of nitrification in recent years. Therefore, owing to the specific fertilization and management methods used in paddy fields, the increase in TN and AHN over the chronosequence promoted an increase in the abundance of *Nitrospirae*. In the fungal communities, most fungal genera were negatively correlated with pH because fungi prefer acidic environments, and the diversity and biomass of soil fungi generally increase with decreasing pH [55]. As one of the most important soil physicochemical properties, SOM affects the soil microbial community structure, microbial biomass, basic respiration, and metabolic entropy and is thus most closely related to the microbial community [56].

Soil characteristic metabolites play important roles in regulating microbial interactions. Long-chain fatty acids are positively correlated with the bacterial genera, and their role in paddy soil should not be ignored. When crops are infected by diseases, their roots secrete high concentrations of long-chain fatty acids to recruit beneficial microbes, which can immediately inhibit pathogens by producing antimicrobial compounds [57]. Short-chain fatty acids can decrease the relative abundance of the *Sphingomonas* and *Pseudomonas* bacterial genera in the infected plant rhizosphere [58]. These findings are consistent with our research. The long-term low-oxygen state of paddy soil inhibited the activity of phenoloxidase, which led to the accumulation of phenolic substances, but the bacteria that evolved for the detoxification or utilization of these compounds can make the substituents on the benzene ring of phenolic acid molecules more prone to biochemical reactions such as decarboxylation, oxidation, and hydroxylation [59]. Therefore, different bacterial genera presented different correlations with phenolic acids. *Woeseia*, *Gaetbulibacter*, and *Seohaeicola* were negatively correlated with phenolic acids, whereas *Defluviicoccus*, *Haliangium*, *Sulfurifustis*, *Sphingomonas*, and *Geobacter* were positively correlated. Although carbohydrates and polyalcohols were negatively correlated with bacteria, they are the most abundant small-molecule metabolites and are important energy sources for microbes in the soil, which can improve the adaptability of microbes under stress conditions. Therefore, these metabolites are highly important in the regulation of microbes in soil [60, 61]. Unlike bacterial genera, fungi had more positive correlations than negative correlations with characteristic metabolites, and fungi can secrete large amounts of organic acids; thus, these metabolites may be derived from fungi [62]. Notably, in this study, only the correlations between microbes and metabolites were explored. In-depth microbial–metabolite dynamic changes and associated mechanisms still need to be studied in detail. The combination of metagenomics, metaproteomics, and metatranscriptomics with soil metabolomics data produced metabolic pathway and signalling molecule findings in this study that will enable a comprehensive understanding of soil microecology.

Conclusions

In this study, we focused on microecological changes during the development of a paddy soil chronosequence on the basis of metabolome and microbiome. The dominant bacterial community showed more regular changes than did the dominant fungal community at the phylum level. Seventy-five characteristic metabolites presented three trends of change with chronosequence. These metabolites were enriched mainly in amino acid metabolism,

energy metabolism and carbohydrate metabolism, which are key carbon and nitrogen metabolic pathways. In addition, we demonstrated that the soil microbial community structure underwent orderly succession under the combined influence of both soil physicochemical properties and metabolites. Except driven by pH, organic matter, available potassium, and total nitrogen, the long-chain fatty acids, short-chain fatty acids, phenolic acids, carbohydrates, and polyalcohols play important roles in regulating microbial interactions. These results revealed that microbial communities and metabolic functions evolved in an orderly manner to increase productivity in the paddy soil chronosequence. We inferred that the close relationship between soil microbes and metabolites can be used to shape the bacterial community structure and function to improve soil quality and increase crop yield.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40538-024-00673-y.

Additional file 1: Figure S1. 53 differentially bacterial abundant taxonomic clades with an LDA score higher than 4.0. **Figure S2.** 35 differentially fungal abundant taxonomic clades with an LDA score higher than 4.0.

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Author contributions
DL performed all experiments. DL and KC designed the study and wrote the majority of the manuscript. WG, ZK, and ALW provided critical comments on the study and helped write the paper. DL, JZ, and DC analyzed the data. DL, KC, and XJ participated in the design of the study, provided comments, and edited the manuscript. All authors read and approved the final manuscript.

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Availability of data and materials
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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